Brain Modelling Project Report

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Note to the reader: This report is intended to accompany the Jupyter notebook submitted as part of the coding project. While the notebook contains all code and plots, this document focuses on summarizing the network design, implementation choices, key findings and other insights we gained. To fully understand the experimental results the report should be read in combination with the notebook file `project.ipynb`. We also want to mention that the submitted notebook represents the final polished version, with many exploratory steps and intermediate insights from experimenting with the network removed to keep it concise and focused. Therefore, some details referenced in the text may not appear explicitly in the notebook but can be readily reproduced if desired.

1 Introduction

The goal of our project was to explore how a biologically inspired spiking neural network, specifically one based on the Izhikevich neuron model, responds to various external stimuli, and how well this response can be used to decode the input using population firing rates. We were particularly interested in how different parameters of the network and the input affect the ability to distinguish between different types of stimuli. To do this, we implemented the network, created several variations of input stimuli, and used a simple Bayesian decoding approach to estimate which stimulus had likely been presented, based on the observed spiking activity.

We decided to study this model due to its placement on the graph 1: it's realistic enough to mimic the interesting firing patterns seen in real neurons, yet simple enough to run quickly on an ordinary laptop.

2 Network Design and Implementation

We used the implementation of the Izhikevic network discussed in class which is based on the original implementation in the paper of Izhikevic [2] and extended it to our needs. After experimenting with different neuron parameters and comparing network behaviour, we narrowed down on using the original implementation. Specifically, we chose to use two types of neurons from the paper as also discussed in class: regular spiking (RS) neurons for excitatory cells, and low-threshold spiking (LTS) neurons for inhibitory cells. These types are defined by different parameters in the Izhikevich model, and we used the same values as in the original paper to stay consistent, also including the method for achieving heterogeneity between the neurons. Exact equations and choice of parameters can be found in their paper [2] and in the notebook.

The total network size was fixed at 1000 neurons, with 800 excitatory and 200 inhibitory neurons. This ratio of 80% excitatory to 20% inhibitory was kept constant throughout the project.

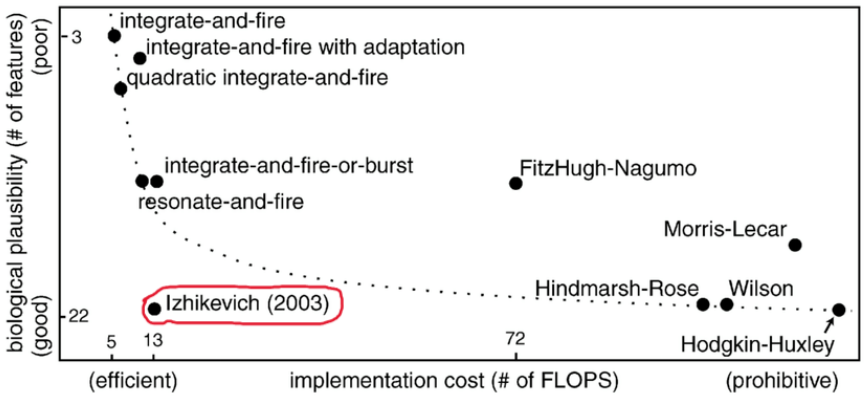


Figure 1: Computational cost versus biological plausibility for several neuron models (redrawn from [1]). The Izhikevich model (highlighted) balances realism and efficiency, making it well-suited for stimulus-decoding studies.

This proportion reflects typical cortical circuits [2]. Initially, we had planned to vary the E/I ratio to see how it affects the network's ability to encode the stimuli. However, during experimentation we found that this was difficult to control in a meaningful way. We initially expected a runaway excitation or complete suppression with an excessive unbalanced ratio. Instead, when we increased or decreased the proportion of excitatory neurons the network continued to behave in fairly regular ways, and the effects of changing the E/I ratio were less pronounced than expected. Usually a higher ratio of excitatory neurons led to better distinguishment. We suspect this is because such effects depend more on specific connectivity and weight patterns than just the overall ratio. Since generating such detailed architectures would have required a lot of fine-tuning and debugging, we decided to stick with the proposed ratio.

Table 1: Default parameters of the simulated Izhikevich network. Values marked "variable" were systematically swept in different experiments.

Symbol	Description	Default value
N_e	Excitatory (RS) neurons	800
N_i	Inhibitory (LTS) neurons	200
N	Total neurons ($N_\mathrm{e} + N_\mathrm{i}$)	1000
T	Simulation time (ms)	1000
a, b, c, d (RS)	Izhikevich parameters for regular spiking	0.02, 0.20, −65, 8
a, b, c, d (LTS)	Izhikevich parameters for low-threshold spiking	0.02, 0.25, −65, 2
w_e	Excitatory synaptic weight	0.60
w_i	Inhibitory synaptic weight	−1.6
p_conn	Connection probability	variable

Another important choice was how we modeled the network's connectivity. Neurons were connected randomly, with the probability of forming a connection being one of the key parameters

we varied. We wanted to understand how denser or sparser connectivity affects the spread of activity through the network and its ability to represent input. From a conceptual point of view, sparse networks might limit signal propagation but reduce interference and noise, while dense networks could improve signal strength but also make the network more chaotic or noisy.

We also looked at how changing the synaptic weights affects the network's behavior. We tested different levels of excitatory and inhibitory weights, again based on values from the paper. These were chosen somewhat arbitrarily within reasonable biological ranges, and our goal was to see how different weight combinations shape the firing patterns and ultimately the ability to decode stimuli.

Throughout the experiments, we kept the noise in the network at a constant level. The noise was generated by random inputs and fluctuations that simulate background activity, and we chose a value that led to realistic, yet stable, behavior. While varying noise could be interesting, we wanted to focus on structural and input-related parameters.

3 Stimulus Design and Motivation

To study how the network responds to different stimuli, we created several types of external inputs. These included simple shapes like steps and ramps, as well as oscillatory patterns like sine and cosine waves clipped to only positive values. In addition, we added two modified versions of the sine wave: one with a higher amplitude and one with a small delay (i.e., phase shift).

The reason for using similar waveforms, especially the three sinusoidal ones, was to test how well the network can distinguish inputs that differ mainly in amplitude, timing, or simple waveform shape. This allowed us to probe whether the network is more sensitive to changes in firing rate or to timing differences. We also kept the maximum amplitude of the stimuli constant across all inputs except the high amplitude sine, to make comparisons fair and to isolate the effects of timing or waveform shape. The actual amplitude values were chosen based on trial and error, they had to be strong enough to cause a visible response in the network, but not so strong that they overwhelmed its internal dynamics.

One of the most interesting aspects for us was how many neurons received the input. This was defined as a percentage of the total excitatory population, and we varied it across several values (e.g., 1%, 5%, 10%). The idea behind this was to understand how much external input is needed to overcome the internal noise and variability of the network. If only a few neurons get input, the signal might get lost or masked by background activity. If many neurons get input, the signal should be more visible and easier to decode. This variation turned out to be quite impactful and helped reveal how network scale and input distribution interact.

4 Bayesian Decoding Method

To analyze the network's response to the stimuli, we used a Bayesian decoding approach. We chose this method because it was covered in class and is relatively simple to implement while still being effective for small datasets. The idea is to use the firing rates from the network to infer which stimulus was presented. For each trial, we computed the firing rate across the whole network in time bins of 10 milliseconds, giving us a time series of 100 data points per trial. We ran a maximum of 40 trials per stimulus, which was a trade-off between simulation time and getting enough data. We then split the trials into training and testing sets. Seventy percent of the trials were used to learn the firing rate distributions for each stimulus. For each time bin, we computed the mean and variance of the firing rate. We assumed that the firing rates followed a normal (Gaussian) distribution. Then, for each test trial, we calculated the likelihood that the

observed firing rate came from each stimulus' distribution. This gave us a probability for each stimulus.
We assume that each stimulus was equally likely before observing any data (prior) and used Bayes' theorem to compute the posterior probabilities. Although the absolute values of the posteriors were very small due to the dimensionality of the data, normalizing them and using log-likelihoods gave us stable and meaningful predictions. The decoded stimulus was the one with the highest posterior probability.
We evaluated the decoding performance using both classification accuracy and ROC curves, particularly the Area Under the Curve (AUC), which gives a more nuanced picture of how well the decoder distinguishes between classes.

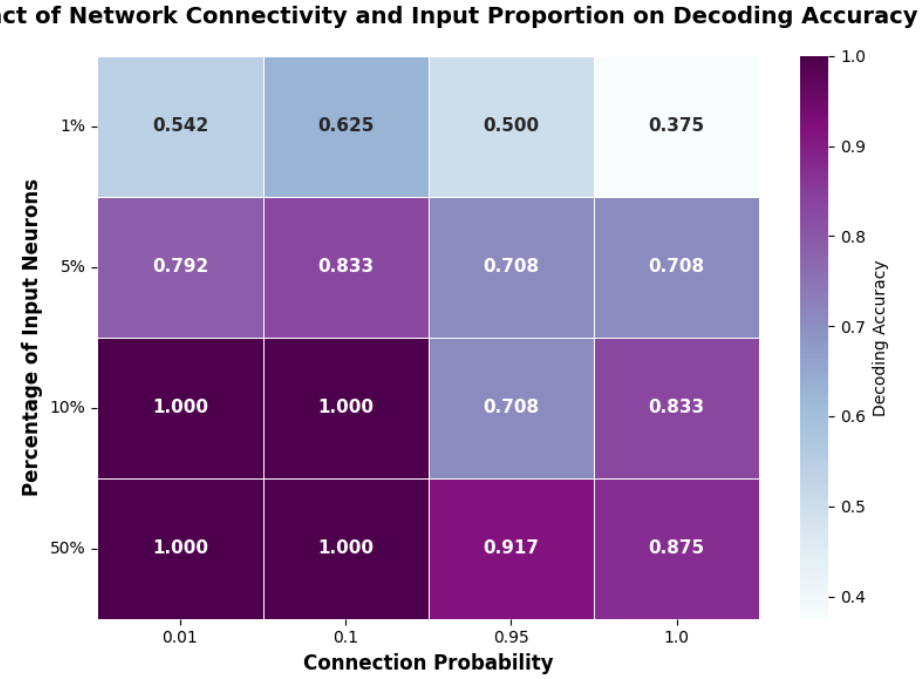


Figure 2: Decoding accuracy for the delayed-sine task as a function of connection probability (columns) and percentage of input neurons (rows). The values inside each cell are the mean accuracies over 20 trials. Darker colours indicate better performance.

5 Results and Observations

The complete results, including detailed plots, can be found in the accompanying notebook. Here we present a summary of the most important findings and their interpretation. First, we tested the network's ability to distinguish between different input stimuli using our Bayesian decoding method. When classifying between multiple stimuli, including several sinusoidal variations, we found an overall classification accuracy of 67.9%. As expected and desired, most errors were among the three sinusoidal variants; a smaller fraction occurred between the ramp and step. This result confirms that the decoder correctly identified the more distinct stimuli and struggled

more with inputs that are only subtly different, such as the delayed sine or the version with slightly higher amplitude but also the ramp and step stimulus.

Further, we analyzed the receiver operating characteristic (ROC) curves for the sinus like stimuli. We found that the ROC curves for the sinusoidal variations were fairly similar, indicating that while the network can distinguish them to some extent, the decision boundary is not sharp. This was intentional in our design: we chose inputs that would be similar enough to challenge the decoder, while still producing distinguishable network responses. This setup created a useful testing ground for exploring how network structure affects decoding performance under difficult conditions.

We also examined how the strength of synaptic weights within the network influenced the Area Under the Curve (AUC) for each classification task. A consistent pattern emerged: networks with weaker synaptic connections tended to produce higher AUC values, meaning better discrimination between stimuli. This makes sense, as weaker internal connections reduce the impact of recurrent noise and make the input-driven component of the activity more dominant. However, if the weights are too weak, the network becomes unresponsive or loses meaningful dynamics. Thus, a balance must be found where the network retains enough internal activity to process and propagate signals, without drowning out the influence of the input.

In the final stage, we narrowed the focus to a particularly subtle classification task: distinguishing a sine wave input from a slightly delayed version of itself. This tested the network's sensitivity to small temporal differences. For these tests, we varied two parameters systematically: the proportion of neurons receiving the input, and the connection probability within the network.

However, the relationship was not strictly linear (see Fig. 2). In some cases, too much connectivity led to erratic behavior, likely due to excessive internal excitation or recurrent feedback. There appeared to be a sweet spot, the range of connectivity and input size where performance peaked. Beyond this range, improvements flattened or even reversed slightly. These findings emphasize that while stronger input and connectivity help up to a point, too much connectivity can lead to reduced performance.

Taken together, the patterns in Fig. 2 reveal the intricate balance required for effective stimulus encoding in spiking networks. Even subtle variations in structure, noise, or input distribution can produce noticeable shifts in decoding accuracy, underlining the sensitivity and complexity of these systems.

6 Discussion and Reflections

The results of our experiments highlight how finely tuned the dynamics of spiking neural networks must be to achieve reliable encoding and decoding of stimuli. One of the key insights from our study is that stimulus discriminability in such networks is highly dependent on the internal architecture as well as the structure and strength of the external input.

From the observed ROC curves and decoding accuracies, we can interpret that the network encodes different aspects of the stimulus depending on its parameters. For instance, in networks with weaker synaptic weights, the external stimulus plays a more dominant role in shaping the firing activity, which benefits decoding performance. Conversely, stronger recurrent connections may introduce more internal variability or noise, making it harder for the decoder to identify clear differences in the response to similar inputs. Our investigation of temporal sensitivity suggests that even biologically simple networks like this can be sensitive to subtle temporal features, provided the input engages a sufficient number of neurons and is allowed to propagate through an appropriately connected architecture. The fact that decoding performance did not improve

indefinitely with denser connectivity also shows that more is not always better. Instead, optimal network behavior arises from a balanced interaction between internal structure and external stimulation.

One surprising outcome was that altering the excitatory/inhibitory ratio, which we initially expected to have a dramatic effect on the network's stability and encoding capacity, had only modest effects in practice. We suspect this is because network behavior depends not just on overall ratios but also on detailed patterns of connectivity and weights. This highlights a potential limitation of our simplified random connectivity model and points to a direction for future work.

There are several open questions and directions that would be interesting to explore in future experiments. For example, it would be valuable to test more biologically realistic connectivity patterns, such as small-world or modular architectures, and to investigate how these affect stimulus encoding. It would also be interesting to explore the effects of different types of noise, or how learning rules like STDP (spike-timing-dependent plasticity) might adapt the network to improve decoding over time. Finally, integrating temporal decoders, such as those based on spike timing rather than just rates, could provide a richer picture of how information is represented in such networks. In general there are many ways to improve our methodology which we felt to be out of the scope of this project.

7 Conclusion

In this project, we explored how various parameters influence the encoding performance of an Izhikevich spiking neural network. We found that the behavior of such a network is highly sensitive to both its internal structure and the properties of the external input, including how many neurons are stimulated and how different the stimuli are.

A central takeaway is that while increasing the amount of input generally improves performance, this improvement eventually plateaus, and excessive connectivity or overly strong recurrent interactions can even degrade decoding accuracy. These findings emphasize that spiking neural networks operate best within carefully balanced regimes, where input-driven dynamics can interact productively with intrinsic activity.

Decoding neural signals is not straightforward. We learned that even with a relatively simple Bayesian approach, decoding performance depends strongly on how the data is structured, how stimulus differences are reflected in network activity, and how noise interacts with the signal. This made the analysis both challenging and insightful. Practically speaking, a large portion of our time went into trial and error: debugging simulation issues, tuning parameters, and interpreting noisy results. The final notebook represents only the most coherent and interpretable subset of these trials. Still, we believe it illustrates the core dynamics and phenomena we set out to explore.

References

[1] Eugene M. Izhikevich. "Which Model to Use for Cortical Spiking Neurons?" In: *IEEE Transactions on Neural Networks* 15.5 (2004), pp. 1063–1070. DOI: 10.1109/TNN.2004.832719.

[2] Eugene M. Izhikevich. "Simple Model of Spiking Neurons". In: *IEEE Transactions on Neural Networks* 14.6 (2003), pp. 1569–1572. DOI: 10.1109/TNN.2003.820440.